

Some Important Questions

Q. No.	Questions
1.	Define the term 'schema'.
2.	State the primary goal of the Normalization process.
3.	Identify the normal form that addresses transitive dependency.
4.	List out the three main levels of Data Abstraction.
5.	Rewrite the relational algebra expression for the Rename operation (ρ).
6.	Classify the different types of Join operations in SQL.
7.	Restate the purpose of the Rollback command.
8.	Write the DML command used to change data in a table.
9.	Give an example of a functional dependency.
10.	Define the term 'transaction'.
11.	State the purpose of Query Processing.
12.	Enlist two disadvantages of the File Processing System.
13.	Define Data Abstraction.
14.	State the types of Relational Integrity Constraints.
15.	Identify the types of Databases.
16.	List out the Data Manipulation Operations in a DBMS.
17.	Rewrite the difference between DDL and DML with examples.
18.	Classify the differences between MySQL and SQL Server.
19.	Restate the significance of normalization in improving database design.
20.	Write the significance of multivalued dependency .
21.	Give example of evaluating a join ($\bowtie$) operation using relational algebra.
22.	Define a selection (σ) operation in relational algebra.
23.	State the four properties of a transaction (ACID) .
24.	Enlist the difference between unique key and primary key .
25.	Define the term DBMS.
26.	State the role of Data Manipulation Language (DML) in a DBMS.
27.	Identify the characteristics that distinguish the database approach with File based approach.
28.	List any two undesirable properties that a database design may have.
29.	Rewrite the advantages of Normalization.
30.	Classify different types of functional dependency
31.	Restate the advantages of relational model.
32.	Write the limitations of a file processing system.
33.	Give one example of DDL.
34.	Define the term SQL.
35.	State the difference between BCNF with 3NF.
36.	Enlist various types of attributes.
37.	Define Data independence.
38.	State the advantages of file processing system
39.	Identify the types of relational algebra operations.
40.	List out the components of E-R Model.
41.	Rewrite SQL.
42.	Classify the types of serializability.
43.	Restate the definition of BCNF.
44.	Write about transitive functional dependency.
45.	Give the states of transaction.
46.	Define Primary index and Secondary Index.
47.	State the need for Query Optimization.
48.	Enlist the properties of decomposition.

49.	Distinguish between Tuple Relational Calculus and Relational Algebra.																								
50.	Illustrate the concept of Transitive Dependency with a non-normalized table example.																								
51.	Discuss the concept of Data Abstraction in a DBMS.																								
52.	Examine the process of using B-trees for efficient data retrieval.																								
53.	Write short note on the ACID Properties of a transaction.																								
54.	Sketch the E-R Diagram for a simplified scenario (e.g., Department and Employee).																								
55.	Distinguish between Strong Entity Set and Weak Entity Set .																								
56.	Illustrate the levels of data abstraction (physical, logical, view).																								
57.	Discuss the role of tuple relational calculus in query formulation.																								
58.	Examine the usefulness of higher normal forms (4NF, 5NF) in reducing anomalies.																								
59.	Write short note on the steps in query processing .																								
60.	Sketch the organization of a B-tree used for indexing.																								
61.	Distinguish between primary key and unique key constraint.																								
62.	Illustrate Group by and Having clauses with examples?																								
63.	Discuss following relational algebra operations with example: a. Natural join operation b. Projection operation																								
64.	Examine three levels of data abstraction in DBMS.																								
65.	Write short note on Decomposition.																								
66.	Sketch the structure and usage of B-trees in database indexing.																								
67.	Distinguish between logical and physical data independence.																								
68.	Illustrate the SQL query for the given table- <table><tr><th>Emp Id</th><th>Emp position</th><th>DOJ</th><th>Salary</th></tr><tr><td>1</td><td>Manager</td><td>01/05/2019</td><td>5000000</td></tr><tr><td>2</td><td>Executive</td><td>02/05/2019</td><td>75000</td></tr><tr><td>3</td><td>Manager</td><td>02/11/2024</td><td>90000</td></tr><tr><td>2</td><td>Lead</td><td>05/06/2021</td><td>85000</td></tr><tr><td>1</td><td>Executive</td><td>01/05/2019</td><td>300000</td></tr></table> a) Write a query to find all the employees whose salary is between 50000 to 100000. b) Write a query to find the details of employee those who have work on lead position.	Emp Id	Emp position	DOJ	Salary	1	Manager	01/05/2019	5000000	2	Executive	02/05/2019	75000	3	Manager	02/11/2024	90000	2	Lead	05/06/2021	85000	1	Executive	01/05/2019	300000
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69.	Discuss Static Hashing and Dynamic Hashing																								
70.	Examine normalization. Explain the need for normalization.																								
71.	Write short note on DML.																								
72.	Sketch with neat diagram 2 tier and 3 tier architecture.																								
73.	Illustrate the working of the Two-Phase Locking (2PL) protocol with a sample transaction schedule.																								
74.	Explain and justify the necessity of the Lossless Design property for database decomposition.																								
75.	Analyze the process of Query Optimization by decomposing a query plan into relational algebra expressions.																								
76.	Examine the various File Organization methods.																								
77.	Describe and critique the Timestamp-based Concurrency Control scheme.																								
78.	Discuss the differences between Relational Algebra and Relational Calculus.																								
79.	Construct a SQL queries for Employee details with attributes Eid, Ename, Salary, Department and location using DDL and DML Command.																								
80.	Explain the limitations of file processing systems and how DBMS overcomes them.																								
81.	Analyse the suitability of MySQL vs Oracle for small-scale vs enterprise-level applications.																								
82.	Examine the Boyce-Codd Normal Form (BCNF) and explain how it differs from 3NF.																								
83.	Describe the role of query equivalence in query optimization .																								
84.	Discuss the different types of indices used in database systems with examples.																								
85.	Illustrate the three level architecture in detail.																								
86.	Consider following schema and write SQL for given statements. Student (rollno, name, branch) Exam (rollno, subject_code, obtained_marks , paper_code) Papers (paper code, paper satter name, university)																								

	(i) Display name of student who got first class in subject '130703'. (ii) Display name of all student with their total mark. (iii) Display list number of student in each university. (iv) Display list of student who has not given any exam
87.	Analyse different types of database users.
88.	Examine Order by, Group by and Having Clauses with example.
89.	Describe transaction processing with its ACID properties in DBMS.
90.	Discuss the following transaction with relevant example: I. Read Only Transaction II. Read Write Transaction III. Aborted Transaction
91.	Illustrate with the help of the diagram, the basic architecture of a database management system.
92.	Explain Order by, Group by and Having Clauses with example.
93.	Analyse about lossy and lossless Decomposition.
94.	Examine in detailed about Query processing.
95.	Describe the ACID Properties of a transaction.
96.	Discuss First normal form, Second normal form and third normal with an example.
97.	Rewrite about the following: a) The concept of Serializability (Conflict and View) The purpose of Database Recovery.
98.	Explain the concepts of Functional Dependencies and the different Normal Forms (2NF, 3NF, BCNF) with a detailed example of normalization.
99.	Describe the full process of Query Processing from parsing to execution, and construct a high-level overview of different Join Strategies.
100.	Illustrate the Entity–Relationship (E-R) model with a detailed diagram for a university scenario.
101.	Explain the requirements and examples of 1NF, 2NF, and 3NF .
102.	Describe the overall architecture of DBMS.
103.	Rewrite about following: a) Hashing. b) Indexing. Concurrency control.
104.	Explain about Query processing and Optimization. Describe the cost estimation of Query Optimization
105.	Describe the DBMS Architecture in detail.
106.	Rewrite about following: a) Data Models. b) Mapping cardinalities.
107.	Explain how will you convert that relation into BCNF. Give an example of a relation that is in 3NF but not in BCNF.
108.	Develop an ER model using step by step representation with all components for a university database application where- a) University has many departments. b) Each department has multiple instructors one among them is the head of department. c) An instructor belongs to only one department. d) Each department offers multiple courses, each of which is taught by a single instructor. A student may enrol for many courses offered by different departments.